

## # Solving Exact DEs:

↳ For: 1st order ODEs

↳ The DE is of the form:

$$M \cdot dx + N \cdot dy = 0$$

and

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Condition for a DE to be exact

← M, N are both functions of x and y

↳ Method:

To solve  $M \cdot dx + N \cdot dy = 0$ ,

- first integrate the terms in  $M \cdot dx$  as if  $y$  were a constant (kind of like partial integration with respect to  $x$ ).
- Then integrate the terms in  $N \cdot dy$  as if  $x$  were a constant (kind of like partial integration with respect to  $y$ ). Reject the terms already obtained from integrating  $M dx$ .
- Equate the sum of the remaining terms to a constant.

(e.g.) Solve:  $(2x^3 + 4y) \cdot dx + (4x + y - 1) \cdot dy = 0$

Soln Let  $M = 2x^3 + 4y$  and  $N = 4x + y - 1$

$$\therefore \frac{\partial M}{\partial y} = 4 \quad \text{and} \quad \frac{\partial N}{\partial x} = 4$$

Since  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ , the DE is exact.

$$\begin{aligned} \rightarrow \int M dx &= \int (2x^3 + 4y) dx \\ &= \frac{2x^4}{4} + 4yx + C \\ &= \frac{x^4}{2} + 4xy + C_1 \end{aligned}$$

$$\begin{aligned} \rightarrow \int N dy &= \int (4x + y - 1) dy \\ &= 4xy + \frac{y^2}{2} - y + C_2 \end{aligned}$$

Hence rejecting the repeated terms and taking the sum of the remaining terms, we equate it to a constant  $C$ :

$$\frac{x^4}{2} + 4xy + \frac{y^2}{2} - y = C$$

$$\Rightarrow x^4 + 8xy + y^2 - 2y = \underline{\underline{C}}$$

★ If a DE is of the form  $Mdx + Ndy = 0$  but is NOT exact, we first convert it to an exact DE using the next method.

## # Method of Integrating Factors:

↳ For: 1st order ODEs

↳ The DE is of the form:

$$M \cdot dx + N \cdot dy = 0$$

and  $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \quad \leftarrow \text{The DE is not exact.}$

↳ We can convert this DE to an exact form.

This is done by multiplying both sides of the DE by something called an integrating factor.

\* An integrating factor (IF) of a non-exact DE is a function, in terms of  $x$  and  $y$ , which can be multiplied to the DE to make it exact.

i.e.  $M dx + N dy \rightarrow \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \quad \leftarrow \text{non-exact}$

but  $M \cdot IF \cdot dx + N \cdot IF \cdot dy \rightarrow \frac{\partial (M \cdot IF)}{\partial y} = \frac{\partial (N \cdot IF)}{\partial x} \quad \leftarrow \text{exact}$

↳ Rules for determining integrating factors:

Rule 1: If  $\left( \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \right)$  is a function of  $x$  only,

say  $f(x)$ , then  $IF = e^{\int f(x) dx}$ .

Rule 2: If  $\left( \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} \right)$  is a function of  $y$  only,

say  $f(y)$ , then  $IF = e^{\int f(y) dy}$ .

Rule 3: If  $M$  and  $N$  are both homogeneous <sup>NOT the same as homogeneous DEs</sup>

functions to each other in  $x$  and  $y$ ,

then  $IF = \frac{1}{Mx + Ny}$  (given  $Mx + Ny \neq 0$ ).

<sup>note that  $x$  and  $y$  are being used here NOT  $dx$  and  $dy$ .</sup>

Rule 4: If the main DE is of the form:

$$y \cdot f(x, y) \cdot dx + x \cdot g(x, y) \cdot dy = 0$$

then  $IF = \frac{1}{Mx - Ny}$  (given  $Mx - Ny \neq 0$ ).

\* A function is said to be homogeneous to another if all the terms in both the functions are of the same combined degree.

e.g.  $f(x, y) = 3x^3 + 5xy^2$

this term has degree 3  
 $x$  has power = 3  
this term also has degree 3  
 $x$  has power = 1  
 $y$  has power = 2  
 $1 + 2 = 3$   
combined degree of  $x$  and  $y$

$$g(x, y) = 2x^2y - y^3$$

this term also has degree 3  
 $x$  has power = 3  
this term has degree 3  
 $x$  has power = 2  
 $y$  has power = 1  
 $2 + 1 = 3$   
combined degree of  $x$  and  $y$

Now, each term in  $f(x, y)$  has degree = 3 and each term in  $g(x, y)$  also has degree = 3.

$\therefore f(x, y)$  is homogeneous to  $g(x, y)$ .

\* To learn more on this, if you're interested, look up "degree of homogeneity of a function" on the internet.

\* When attempting Rule 4, you can't forcefully bring the DE into the required form if it's not already in the said form.

e.g.  $M dx + N dy = 0$

$$\Rightarrow 4xy dx + 5y^2 dy = 0$$

Here  $M$  can be written as  $4xy = y(4x) = y \cdot f(x,y)$   
but we can't bring  $N = 5y^2$  into the form  $x \cdot g(x,y)$ .  
You might try:

$$N = 5y^2 = x \cdot \left(\frac{5y^2}{x}\right)$$

but that would be **wrong** as  $x$  was not initially present in  $N$ .

(e.g.) Solve:  $(2x^2 + y^2 + x) dx + (xy) dy = 0$

Sol<sup>n</sup>  $M = 2x^2 + y^2 + x$   
 $N = xy$

$$\begin{array}{ccc} \xrightarrow{\quad} \frac{\partial M}{\partial y} = 2y & \xrightarrow{\quad} \therefore \left( \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \right) \\ \frac{\partial N}{\partial x} = y & = \frac{2y - y}{xy} = \frac{1}{x} \end{array}$$

$$\therefore IF = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

So multiplying the DE by the IF, we get:

$$(2x^3 + xy^2 + x^2) dx + (x^2y) dy = 0$$

Now  $M_1 = 2x^3 + xy^2 + x^2$

$$N_1 = x^2y$$

$$\therefore \int M_1 dx = \int (2x^3 + xy^2 + x^2) dx = \frac{x^4}{2} + \frac{(xy)^2}{2} + \frac{x^3}{3} + C_1$$

$$\int N_1 dy = \int x^2y dy = \frac{(xy)^2}{2} + C_2$$

$$\therefore \text{DE is: } \frac{x^4}{2} + \frac{(xy)^2}{2} + \frac{x^3}{3} = C_3$$

$$\Rightarrow 3x^4 + 3(xy)^2 + 2x^3 = C //$$

(e.g.) Solve  $(x^3 + y^3) dx - (xy^2) dy = 0$

$$\begin{aligned} \underline{\text{Sol}^n}: \quad M &= x^3 + y^3 \\ N &= -xy^2 \end{aligned} \quad \xrightarrow{\quad} \quad \begin{aligned} \frac{\partial M}{\partial y} &= 3y^2 \\ \frac{\partial N}{\partial x} &= -y^2 \end{aligned} \quad \xrightarrow{\quad} \quad \begin{aligned} \therefore \left( \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \right) \\ &= \frac{3y^2 + y^2}{-xy^2} \\ &= \frac{-4}{x} \end{aligned}$$

$$\therefore \text{IF} = e^{\int \frac{-4}{x} dx} = e^{-4 \ln x}$$

$$= e^{\ln x^{-4}} = x^{-4} = \frac{1}{x^4} \quad \leftarrow \text{using Rule 1}$$

Also, using Rule 3:  $\text{IF} = \frac{1}{M_x + N_y} = \frac{1}{x^4 + xy^3 - xy^3} = \frac{1}{x^4} \quad \leftarrow \text{faster}$

Now, multiplying both sides of the DE by  $\frac{1}{x^4}$ :

$$\left( \frac{1}{x} + \frac{y^3}{x^4} \right) dx - \left( \frac{y^2}{x^3} \right) dy = 0$$

$$\therefore \int M_1 dx = \int \frac{1}{x} + \frac{y^3}{x^4} dx = \ln x - \frac{y^3}{3x^3} + C_1$$

$$\int N_1 dy = \int -\frac{y^2}{x^3} dy = -\frac{y^3}{3x^3} + C_2$$

Hence, the solution to the DE is:

$$\ln x - \frac{y^3}{3x^3} = C //$$

(e.g.) Solve:  $y(1+xy) dx + x(1-xy) dy = 0$

Sol<sup>n</sup>: Here the DE is of the form:

$$y \cdot f(x,y) dx + x \cdot g(x,y) dy = 0$$

$\therefore$  Using Rule 4:

$$\begin{aligned} \text{IF} &= \frac{1}{Mx - Ny} = \frac{1}{xy + x^2y^2 - xy + x^2y^2} \\ &= \frac{1}{2x^2y^2} \end{aligned}$$

Multiplying the DE by the IF:

$$\frac{y + xy^2}{2x^2y^2} dx + \frac{x - x^2y}{2x^2y^2} dy = 0$$

$$\therefore M_1 = \frac{1}{2x^2y} + \frac{1}{2x} \quad N_1 = \frac{1}{2xy^2} - \frac{1}{2y}$$

$$\text{Hence, } \int M_1 dx = \int \frac{1}{2x^2y} + \frac{1}{2x} dx = \frac{-1}{2xy} + \frac{1}{2} \ln x + C_1$$

$$\int N_1 dy = \int \frac{1}{2xy^2} - \frac{1}{2y} dy = \frac{-1}{2xy} - \frac{1}{2} \ln y + C_2$$

$\therefore$  Solution to the DE:

$$\frac{-1}{2xy} + \frac{1}{2} \ln x - \frac{1}{2} \ln y = C$$

$$\Rightarrow \frac{-1}{2xy} + \ln \sqrt{\frac{x}{y}} = C //$$

\* Rules 3 and 4 might be faster, but not always applicable.  
At your level, all the problems you encounter can be solved using Rules 1 and 2.